

R Notebook

CHL5222 Final Project

Breastfeeding and HIV exposure on infant growth

Main outcome: weight-for-length z-score (zwfl)

Secondary outcome: head circumference z-score (zhc)

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```
library(splines)
```

```
library(emmeans)
```

```
## Welcome to emmeans.  
## Caution: You lose important information if you filter this package's results.  
## See '? untidy'
```

```
library(ggeffects)  
library(broom.mixed)
```

```
## Warning: package 'broom.mixed' was built under R version 4.5.2
```

```
library(patchwork)  
library(janitor)
```

```
##  
## Attaching package: 'janitor'  
##  
## The following objects are masked from 'package:stats':  
##  
##     chisq.test, fisher.test
```

```
library(naniar)
```

Import data

```
dat <- read.csv("hiv.csv", stringsAsFactors = FALSE) |>  
  clean_names()
```

Variable Descriptions

The dataset contains repeated growth measurements of infants followed from birth to 24 months to evaluate the relationship between HIV exposure, breastfeeding, and infant growth.

- **newid**
Unique identifier for each infant. Used to link repeated observations from the same child in longitudinal analysis.
- **visit**
Study visit number when the measurement was collected.
- **infage_month**
Infant age in months at the time of measurement. This variable represents the time dimension in the longitudinal analysis.
- **zwfl**
Weight-for-length z-score. This standardized measure compares an infant's weight relative to their length based on WHO growth standards and is used as the primary growth outcome.
- **zhc**
Head circumference z-score. This standardized measure reflects head growth relative to age and is considered a secondary growth outcome.
- **enrol_hiv_status_cat**
Maternal HIV status at enrollment. This variable defines whether the infant was exposed to HIV during pregnancy (CHEU) or not exposed (CHU).
- **ebf_updated**
Exclusive breastfeeding status. Indicates whether the infant was exclusively breastfed.
- **bf_current_1year**
Current breastfeeding status. Indicates whether the infant is currently being breastfed at the time of

observation.

- **b1_sex**
Infant sex.
- **gestage_final**
Gestational age at birth (in weeks). This variable accounts for differences in growth related to prematurity.
- **momage_cat**
Maternal age category at the time of delivery.
- **educ_cat_n**
Maternal education level. This variable represents socioeconomic and demographic characteristics that may influence infant growth.

Inspect data

```
str(dat)
```

```
## 'data.frame':    2934 obs. of  12 variables:
## $ newid          : int  96 96 96 96 96 96 96 96 96 97 ...
## $ visit          : chr  "delivery" "week-3" "week-6" "month-12" ...
## $ enrol_hiv_status_cat: chr  "Negative" "Negative" "Negative" "Negative" ...
## $ momage_cat      : chr  "26 to 35" "26 to 35" "26 to 35" "26 to 35" ...
## $ educ_cat_n      : chr  "Primary or below" "Primary or below" "Primary or below" "Primary or below" ...
## $ b1_sex          : chr  "female" "female" "female" "female" ...
## $ gestage_final    : num  38 38 38 38 38 ...
## $ ebf_updated      : chr  "yes" "yes" "yes" "yes" ...
## $ bf_current_1year : chr  "y" "y" "y" "y" ...
## $ infage_month     : num  0.13 0.81 1.39 11.9 18.13 ...
## $ zwfl            : num  -0.68 -0.22 0.02 1.42 0.98 0.47 0.64 0.46 0.61 0.1 ...
## $ zhc             : num  -0.22 1.88 2.04 0.5 0.79 0.51 1.58 1.11 0.81 2.9 ...
```

```
summary(dat)
```

```
##      newid      visit      enrol_hiv_status_cat      momage_cat
## Min.   : 96.0   Length:2934      Length:2934      Length:2934
## 1st Qu.:179.0   Class :character      Class :character      Class :character
## Median :263.0   Mode  :character      Mode  :character      Mode  :character
## Mean   :262.2
## 3rd Qu.:345.0
## Max.   :428.0
##
##      educ_cat_n      b1_sex      gestage_final      ebf_updated
## Length:2934      Length:2934      Min.   :31.61      Length:2934
## Class :character      Class :character      1st Qu.:38.00      Class :character
## Mode  :character      Mode  :character      Median :39.00      Mode  :character
##                               Mean   :39.09
##                               3rd Qu.:40.00
##                               Max.   :48.48
##
##      bf_current_1year      infage_month      zwfl      zhc
## Length:2934      Min.   : 0.000      Min.   : -5.5600      Min.   : -6.6400
## Class :character      1st Qu.: 1.370      1st Qu.: -0.5900      1st Qu.: -0.7125
## Mode  :character      Median : 6.000      Median : 0.1800      Median : -0.0200
##                               Mean   : 8.262      Mean   : 0.1964      Mean   : -0.0423
```

```
##          3rd Qu.:12.030    3rd Qu.: 1.0100    3rd Qu.: 0.6325
##          Max.      :35.550    Max.      : 5.4800    Max.      : 4.3900
##                                     NA's      :37      NA's      :2
```

```
colSums(is.na(dat))
```

```
##          newid          visit enrol_hiv_status_cat
##          0          0          0
##          momage_cat      educ_cat_n          b1_sex
##          0          0          0
##          gestage_final    ebf_updated    bf_current_1year
##          0          5          22
##          infage_month      zwfl          zhc
##          0          37          2
```

Recode variables

```
analysis_dat <- dat |>
  mutate(
    newid = factor(newid),

    # HIV exposure group:
    # Assuming "Positive" = infant exposed to maternal HIV (CHEU)
    # and "Negative" = HIV-unexposed (CHU)
    hiv_group = case_when(
      enrol_hiv_status_cat == "Positive" ~ "CHEU",
      enrol_hiv_status_cat == "Negative" ~ "CHU",
      TRUE ~ NA_character_
    ),
    hiv_group = factor(hiv_group, levels = c("CHU", "CHEU")),

    # Order visits chronologically
    visit = factor(visit, levels = c(
      "delivery",
      "week-3",
      "week-6",
      "month-3",
      "month-6",
      "month-9",
      "month-12",
      "month-18",
      "month-24"
    )),

    # Sex
    sex = factor(b1_sex),

    # Maternal age category
    momage_cat = factor(momage_cat),

    # Maternal education
    educ_cat_n = factor(educ_cat_n),

    # Breastfeeding variables
```

```

ebf_updated = case_when(
  tolower(ebf_updated) %in% c("yes", "y") ~ "Yes",
  tolower(ebf_updated) %in% c("no", "n") ~ "No",
  TRUE ~ NA_character_
),
ebf_updated = factor(ebf_updated, levels = c("No", "Yes")),

bf_current_1year = case_when(
  tolower(bf_current_1year) %in% c("yes", "y") ~ "Yes",
  tolower(bf_current_1year) %in% c("no", "n") ~ "No",
  TRUE ~ NA_character_
),
bf_current_1year = factor(bf_current_1year, levels = c("No", "Yes")),

# Continuous variables
infage_month = as.numeric(infage_month),
gestage_final = as.numeric(gestage_final),
zwfl = as.numeric(zwfl),
zhc = as.numeric(zhc)
)

```

Basic checks

```

# Number of infants and observations
analysis_dat |>
  summarise(
    n_obs = n(),
    n_infants = n_distinct(newid)
  )

##   n_obs n_infants
## 1  2934         333

# Outcome missingness
analysis_dat |>
  summarise(
    missing_zwfl = sum(is.na(zwfl)),
    missing_zhc = sum(is.na(zhc))
  )

##   missing_zwfl missing_zhc
## 1           37           2

# Group counts
table(analysis_dat$hiv_group, useNA = "ifany")

##
##  CHU CHEU
## 1431 1503

table(analysis_dat$ebf_updated, useNA = "ifany")

##
##   No  Yes <NA>
## 1301 1628     5

```

```
table(analysis_dat$bf_current_1year, useNA = "ifany")
```

```
##
##    No  Yes <NA>
##  330 2582   22
```

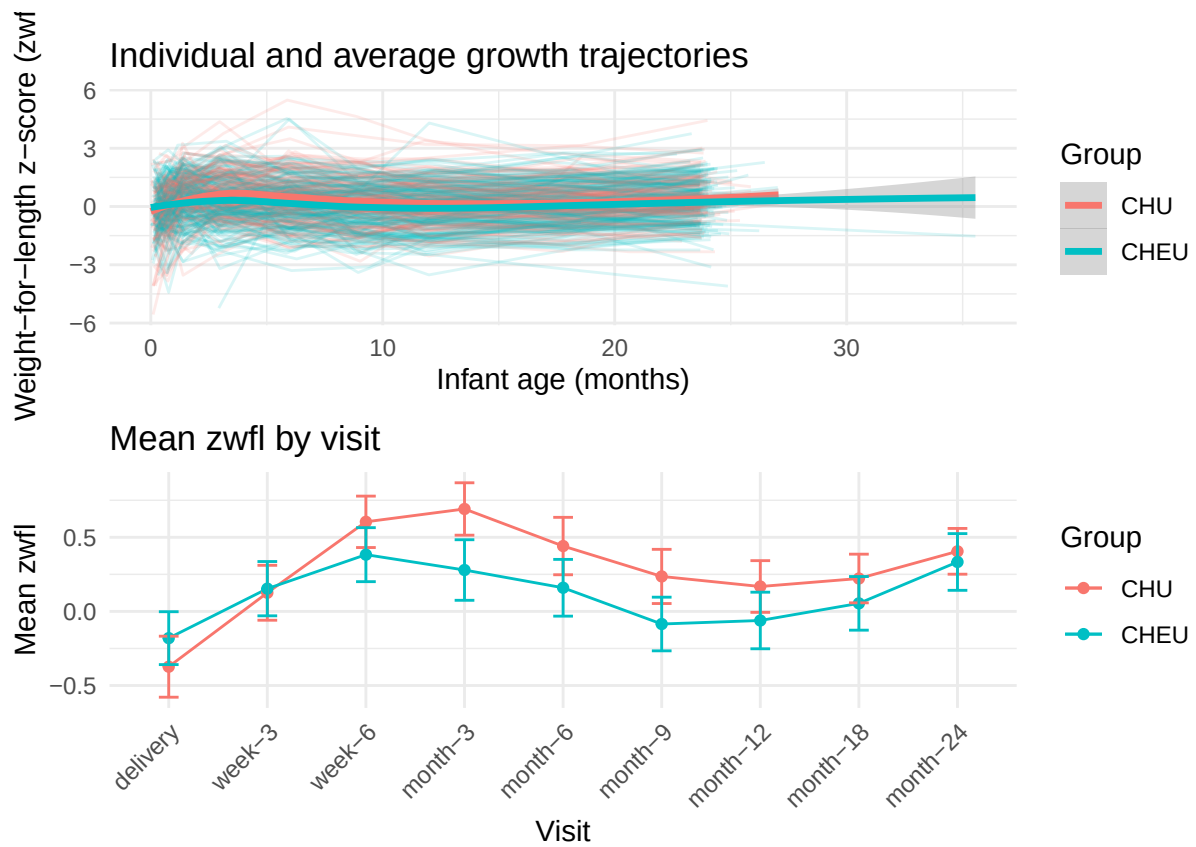
Exploratory plots

```
# Spaghetti plot for zwfl
p1 <- analysis_dat |>
  filter(!is.na(zwfl), !is.na(hiv_group)) |>
  ggplot(aes(x = infage_month, y = zwfl, group = newid, color = hiv_group)) +
  geom_line(alpha = 0.15) +
  geom_smooth(aes(group = hiv_group), method = "loess", se = TRUE, linewidth = 1.2) +
  labs(
    title = "Individual and average growth trajectories",
    x = "Infant age (months)",
    y = "Weight-for-length z-score (zwfl)",
    color = "Group"
  ) +
  theme_minimal()

# Mean profile by visit
p2 <- analysis_dat |>
  filter(!is.na(zwfl), !is.na(hiv_group)) |>
  group_by(visit, hiv_group) |>
  summarise(
    mean_zwfl = mean(zwfl, na.rm = TRUE),
    se_zwfl = sd(zwfl, na.rm = TRUE) / sqrt(sum(!is.na(zwfl))),
    .groups = "drop"
  ) |>
  ggplot(aes(x = visit, y = mean_zwfl, group = hiv_group, color = hiv_group)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = mean_zwfl - 1.96 * se_zwfl,
                    ymax = mean_zwfl + 1.96 * se_zwfl,
                    width = 0.2) +
  labs(
    title = "Mean zwfl by visit",
    x = "Visit",
    y = "Mean zwfl",
    color = "Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

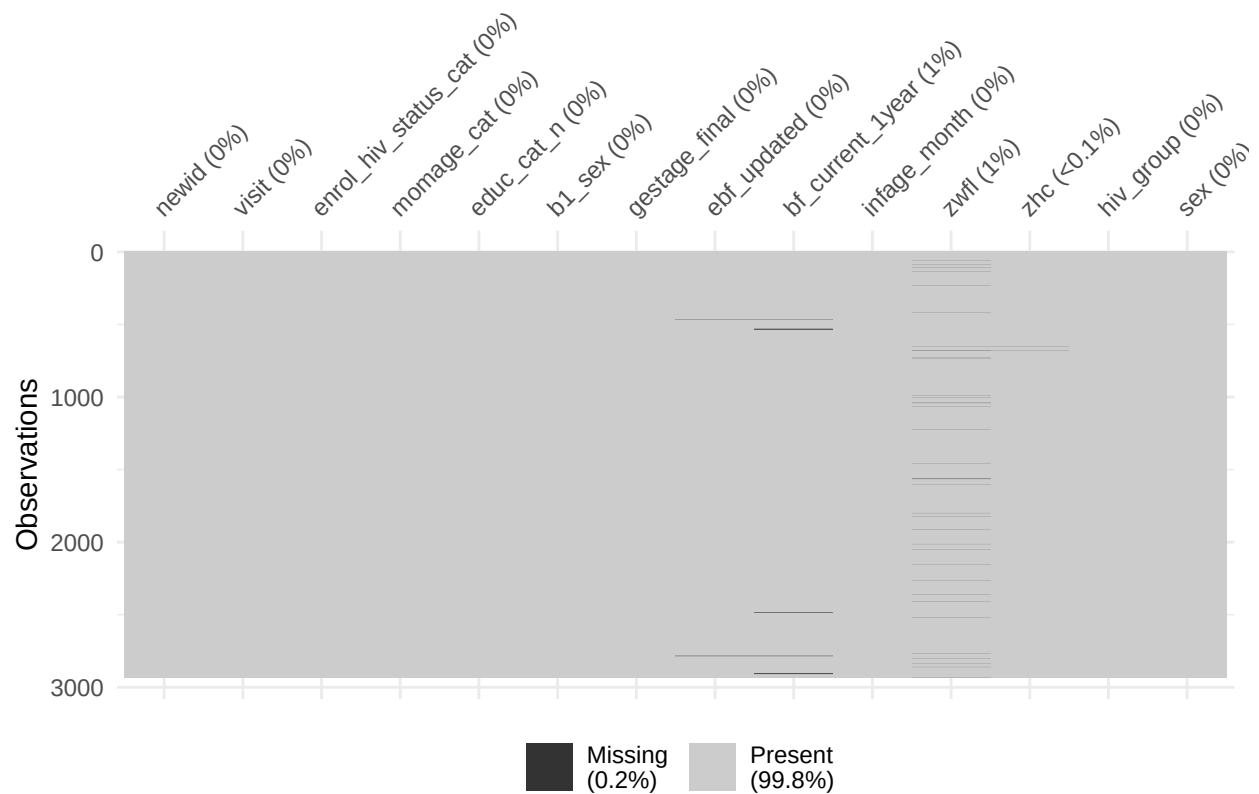
p1 / p2

## `geom_smooth()` using formula = 'y ~ x'
```



Missingness

```
vis_miss(analysis_dat)
```



Main analysis dataset

```
# Keep observations needed for main model
main_dat <- analysis_dat |>
  filter(
    !is.na(zwfl),
    !is.na(hiv_group),
    !is.na(infage_month),
    !is.na(bf_current_1year),
    !is.na(ebf_updated),
    !is.na(sex)
  )
```

Model

```
# Model 1
model1_zwfl <- lmer(
  zwfl ~ hiv_group * infage_month +
    bf_current_1year * infage_month +
    hiv_group * bf_current_1year +
    sex + (1 | newid),
  data = main_dat,
  REML = FALSE
)
```



```

# Model 2
model2_zwfl <- lmer(
  zwfl ~ hiv_group * bf_current_1year * infage_month +
    sex + (1 | newid),
  data = main_dat,
  REML = FALSE
)

summary(model1_zwfl)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: zwfl ~ hiv_group * infage_month + bf_current_1year * infage_month +
## hiv_group * bf_current_1year + sex + (1 | newid)
## Data: main_dat
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##      8304.5      8364.2    -4142.3    8284.5      2865
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.5704 -0.5504  0.0155  0.5927  4.3937
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## newid    (Intercept)  0.6156    0.7846
## Residual                    0.8304    0.9112
## Number of obs: 2875, groups: newid, 328
##
## Fixed effects:
##              Estimate Std. Error      df t value
## (Intercept)    -2.455e-01  2.478e-01  3.678e+02  -0.991
## hiv_groupCHEU   -1.130e-01  2.928e-01  3.376e+02  -0.386
## infage_month     2.776e-02  7.000e-03  2.549e+03   3.966
## bf_current_1yearYes  6.013e-01  2.506e-01  3.650e+02   2.399
## sexmale        -1.216e-01  9.344e-02  3.282e+02  -1.301
## hiv_groupCHEU:infage_month -3.761e-03  4.352e-03  2.553e+03  -0.864
## infage_month:bf_current_1yearYes -2.443e-02  6.840e-03  2.549e+03  -3.571
## hiv_groupCHEU:bf_current_1yearYes -3.245e-03  3.068e-01  3.273e+02  -0.011
##
##              Pr(>|t|)
## (Intercept)    0.322339
## hiv_groupCHEU    0.699853
## infage_month     7.5e-05 ***
## bf_current_1yearYes  0.016948 *
## sexmale         0.194114
## hiv_groupCHEU:infage_month  0.387547
## infage_month:bf_current_1yearYes  0.000362 ***
## hiv_groupCHEU:bf_current_1yearYes  0.991568
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) hv_CHEU infg_m bf__1Y sexmal hv_CHEU:_ i:_ _1

```

```
## hiv_grpCHEU -0.777
## infage_mnth -0.239 0.054
## bf_crrnt_1Y -0.935 0.749 0.208
## sexmale -0.231 0.050 -0.001 0.028
## hv_grCHEU:_ 0.097 -0.125 -0.403 -0.024 -0.001
## infg_m:__1Y 0.214 -0.015 -0.897 -0.231 0.003 0.098
## h_CHEU:__1Y 0.730 -0.939 -0.004 -0.775 -0.047 0.002 0.003
```

```
anova(model1_zwfl)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
## hiv_group      0.4386  0.4386     1   364.80  0.5282 0.4678170
## infage_month   13.2843 13.2843     1  2549.10 15.9980 6.521e-05
## bf_current_1year 11.1520 11.1520     1   424.25 13.4301 0.0002790
## sex            1.4058  1.4058     1   328.19  1.6930 0.1941142
## hiv_group:infage_month 0.6202  0.6202     1  2552.81  0.7469 0.3875471
## infage_month:bf_current_1year 10.5908 10.5908     1  2549.46 12.7543 0.0003618
## hiv_group:bf_current_1year 0.0001  0.0001     1   327.27  0.0001 0.9915678
##
## hiv_group
## infage_month      ***
## bf_current_1year   ***
## sex
## hiv_group:infage_month
## infage_month:bf_current_1year ***
## hiv_group:bf_current_1year
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model2_zwfl)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: zwfl ~ hiv_group * bf_current_1year * infage_month + sex + (1 |
## newid)
## Data: main_dat
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  8305.2   8370.8   -4141.6    8283.2     2864
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.4852 -0.5482  0.0132  0.5895  4.3903
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## newid    (Intercept)  0.6157    0.7847
## Residual                0.8299    0.9110
## Number of obs: 2875, groups: newid, 328
##
## Fixed effects:
##
##              Estimate Std. Error    df
## (Intercept) -0.32595    0.25739 427.18941
## hiv_groupCHEU 0.01038    0.31168 432.21931
```

```
## bf_current_1yearYes          0.68879    0.26187  433.59878
## infage_month                0.03726    0.01080 2549.32563
## sexmale                     -0.12171    0.09345  328.17961
## hiv_groupCHEU:bf_current_1yearYes -0.14082    0.32917  432.10785
## hiv_groupCHEU:infage_month    -0.01840    0.01341 2548.99686
## bf_current_1yearYes:infage_month -0.03477    0.01127 2549.36234
## hiv_groupCHEU:bf_current_1yearYes:infage_month 0.01637    0.01418 2549.44076
##                               t value Pr(>|t|)
## (Intercept)                 -1.266 0.206079
## hiv_groupCHEU                0.033 0.973450
## bf_current_1yearYes          2.630 0.008837 **
## infage_month                 3.450 0.000569 ***
## sexmale                     -1.302 0.193684
## hiv_groupCHEU:bf_current_1yearYes -0.428 0.669016
## hiv_groupCHEU:infage_month    -1.372 0.170122
## bf_current_1yearYes:infage_month -3.085 0.002058 **
## hiv_groupCHEU:bf_current_1yearYes:infage_month 1.154 0.248517
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) hv_CHEU bf__1Y infg_m sexmal hv_CHEU:__1Y hv_CHEU:_ b__1Y:
## hiv_grpCHEU  -0.795
## bf_crrnt_1Y  -0.940  0.773
## infage_mnth  -0.355  0.294  0.350
## sexmale      -0.222  0.047  0.027 -0.002
## hv_CHEU:__1Y  0.753 -0.947 -0.796 -0.278 -0.043
## hv_grCHEU:_   0.286 -0.362 -0.281 -0.805  0.001  0.343
## bf_crr_1Y:_   0.340 -0.281 -0.364 -0.958  0.003  0.290      0.771
## h_CHEU:__1Y: -0.271  0.343  0.290  0.762 -0.001 -0.362      -0.946      -0.795
```

```
anova(model2_zwfl)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                               Sum Sq Mean Sq NumDF    DenDF F value
## hiv_group                    0.1104  0.1104     1    432.13  0.1330
## bf_current_1year             11.7379 11.7379     1    432.39 14.1433
## infage_month                 14.3882 14.3882     1   2549.45 17.3367
## sex                          1.4078  1.4078     1    328.18  1.6963
## hiv_group:bf_current_1year    0.1519  0.1519     1    432.11  0.1830
## hiv_group:infage_month        1.7252  1.7252     1   2549.45  2.0788
## bf_current_1year:infage_month 11.6720 11.6720     1   2549.40 14.0639
## hiv_group:bf_current_1year:infage_month 1.1057  1.1057     1   2549.44  1.3322
##                               Pr(>F)
## hiv_group                    0.7155185
## bf_current_1year             0.0001927 ***
## infage_month                 3.235e-05 ***
## sex                          0.1936837
## hiv_group:bf_current_1year    0.6690160
## hiv_group:infage_month        0.1494813
## bf_current_1year:infage_month 0.0001806 ***
## hiv_group:bf_current_1year:infage_month 0.2485167
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model1_zwfl, model2_zwfl)
```

```
## Data: main_dat
```

```
## Models:
```

```
## model1_zwfl: zwfl ~ hiv_group * infage_month + bf_current_1year * infage_month + hiv_group * bf_curr
```

```
## model2_zwfl: zwfl ~ hiv_group * bf_current_1year * infage_month + sex + (1 | newid)
```

```
##          npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
```

```
## model1_zwfl    10 8304.5 8364.2 -4142.3    8284.5
```

```
## model2_zwfl    11 8305.2 8370.8 -4141.6    8283.2 1.3319  1    0.2485
```